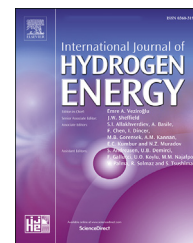


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Thermochemical recuperation by ethanol steam reforming: Thermodynamic analysis and heat balance

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HIGHLIGHTS

- Thermochemical recuperation by steam ethanol reforming.
- Increase in the steam-to-ethanol ratio leads to increase heat deficit.
- Heat deficit in the TCR system with steam-to-ethanol ratio of 3 is from 1 to 2 MJ per 1 kg of ethanol.
- The maximum recovery rate at steam-to-ethanol ratio of 1.

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ABSTRACT

This article considers the scheme of fuel-consuming equipment with a thermochemical heat recuperation system by using ethanol steam reforming. The main concept of thermochemical recuperation (TCR) is the transformation of exhaust gases heat into chemical energy of a new synthetic fuel that has higher calorimetric properties such as low-heating value. Thermochemical recuperation can be considered as an on-board hydrogen production technology. To determine the efficiency of the thermochemical recuperation system, the thermodynamic analysis via Gibbs free energy minimization method was performed. The software Aspen-HYSYS was used for the thermodynamic analysis. The heat flows were calculated for a wide temperature range from 500 to 1000 K, for steam-to-ethanol ratio from 1 to 3, and for various pressures of 1, 5 and 10 bar. The results of the thermodynamic analysis were compared with the experimental results and the results of the thermodynamic analysis performed by other authors. All obtained results are in a good correlation. In the first law energy analysis was found that for a high steam-to-ethanol ratio (above 3), to perform thermochemical recuperation an external heat must be supplied to the TCR system. The heat deficit for steam-to-ethanol ratio 3 is from 1 to 2 MJ/kg_{EtOH} in the temperature range from 500 to 1000 K.

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Introduction

According to the report of the International Energy Agency, by 2030, the use of ethanol in various industries will double in

comparison with 2017 [1]. Nowadays ethanol is widely used in transport and energy as well as in the chemical industry [2,3]. A widespread interest of researchers and scientists is the use of ethanol as a fuel for various fuel-consuming equipment

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such as industrial furnaces [4], gas turbines [5], steam generators [6], etc.

Among the main advantages of using ethanol as a fuel are the following aspects [7]: ethanol is a renewable energy source, good environmental properties, ethanol production, and use do not contribute to the global release of CO₂, if it is properly organized. The main disadvantages of using ethanol in fuel-consuming equipment include the following aspects: the need to change the fuel cycle of existing equipment, a low heating value of ethanol (1 kg produces ≈ 30 MJ [8], while 1 kg of methane produces ≈ 50 MJ), etc.

One way to get around the problem of ethanol using in the existing fuel-consuming equipment is the endothermic transformation of ethanol before using as the main fuel [9]. Such an endothermic transformation can be carried out in the thermochemical heat recuperation systems [10,11]. The main idea of the thermochemical heat recuperation process is to use the flue gases heat for the endothermic transformation of the original fuel. As a result, the transformation products have a higher low heat value (LHV) compared to LHV of the original fuel, in particular, ethanol. Various fuels can be used for thermochemical heat recuperation, for example, methane [12], propane [13], methanol [14], ethanol [15], diesel [16], etc. The work on thermochemical heat recuperation by reforming of liquid fuels is mainly focused on the use in internal combustion engines (ICE) [9,16]. Thermochemical recuperation can be considered as an on-board hydrogen production technology because hydrogen-rich gas fuel is produced in the thermochemical recuperation system.

It is well known that the idea of improving the efficiency of an ICE by the endothermic transformation of initial fuel was discussed as early as in 1960s [17]. The main aim was focused on reducing pollutant emissions when a product gas (gas after endothermic transformation) was used the main fuel in comparison with initial liquid fuel. The historical retrospective of the use of the endothermic transformation in the ICE is comprehensively presented by Tartakovsky and Steintuch [9]. Moreover, there are a set of articles presented by Tartakovsky and co-authors, where the results of using TCR by alcohols reforming are reported [2,14,18]. There is an article discusses the concept of a direct-injection ICE (internal combustion engine) with thermochemical heat recuperation by methanol steam reforming [14]. Poran and Tartakovsky reported that a possibility of efficiency improvement of ICE by 12–14% compared with a DI-gasoline-fueled counterpart by applying a DI-ICE with thermochemical recuperation by methanol reforming at high pressure.

The experimental investigation presented by Ivanic and co-authors shows a slight increase in work output with only 30% fuel conversion [19]. Pratapas and co-workers present a simple engine-TCR system that, when theoretically optimized, would increase efficiency by 10% over a case with no reformation [20]. This increase is surprising because it seems to result from an increase in the peak temperature of only 50 K and a 30% decrease in the peak pressure. It is also not clear what parameters were varied in the optimization process.

A detailed thermodynamic analysis of thermochemical recuperation (TCR) applied to an idealized internal combustion engine with single-stage work extraction was presented by Chakravarthy et al. [21]. The authors reported that for a

stoichiometric mixture of methanol and air, the using of thermochemical heat recuperation can increase the second law efficiency by about 3% for constant pressure reforming and over 5% for constant volume reforming. Also, they reported that the estimated second law efficiency increases for constant volume reforming are 9 and 11% for ethanol and isooctane reforming, respectively.

Moreover, when using thermochemical heat recuperation, an increase in the efficiency of the fuel-consuming equipment can occur not only due to an increase in the energy efficiency but also due to an improvement in the combustion of the fuel. Therefore reducing combustion irreversibility and reducing pollutant emissions may also be considered as the benefits for future implementations of the thermochemical heat recuperation [22].

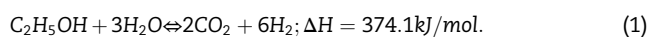
Research studies on the use of TCR by ethanol reforming are mainly focused on thermodynamic, simulation and experimental analysis. However, according to the author, not enough attention is paid to the issues of energy efficiency and heat balance of the thermochemical recuperation systems. Therefore, the aim of this work is to compile the heat balance of the thermochemical recuperation system by steam ethanol reforming. Determining the output and output heat fluxes for the main elements of the TCR system, as well as determining the heat recuperation rate and operating parameters for optimal operation of the thermochemical heat recuperation system.

Analytical approach

TCR concept

A schematic diagram of the fuel-consuming equipment with thermochemical recuperation by steam ethanol reforming is shown in Fig. 1. The schematic diagram can be virtually divided into two blocks: (1) the thermochemical recuperation system (ethanol reformer, steam generator (boiler), ethanol evaporator, mixer (steam ejector)); (2) the fuel consuming system (fuel-consuming equipment itself, burner assembly).

In this schematic diagram, the exhaust gas after fuel-consuming equipment is sequentially passing through the ethanol steam reformer [23], the boiler (steam generator) [24] and the ethanol evaporator. The steam ethanol reformer performs as a catalytic heat exchanger. The reaction mixture before the reformer is containing steam and ethanol vapor. Inside the reaction space of the reformer, ethanol and steam are reacted and to the mixture than mainly consist of hydrogen is produced. The overall chemical endothermic reaction of the steam ethanol reforming process can be written as follows:



The chemical reactions of the ethanol steam reforming process involve the elementary steam ethanol reaction, the water-gas-shift reaction and other side reactions that presented in Table 1 [2,25]. The mechanism of these reactions is depending on operation parameters such as pressure, temperature and steam-to-ethanol ratio, as well as on the catalyst

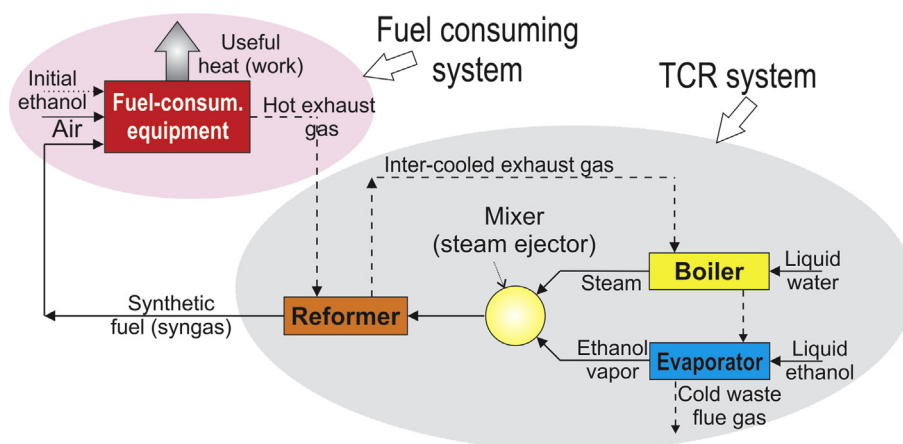


Fig. 1 – Schematic view the fuel-consuming equipment with thermochemical recuperation by steam ethanol reforming.

Table 1 – Possible chemical reactions in the reformer [33].

No.	Reaction	No.	Reaction
The ethanol reforming reaction		Water gas shift reaction	
1	$C_2H_5OH+3H_2O \rightleftharpoons 2CO_2 + 6H_2$	7	$CO + H_2O \rightleftharpoons CO_2 + H_2$
2	$C_2H_5OH + H_2O \rightleftharpoons 2CO + 4H_2$	Carbon formation	
3	$C_2H_5OH \rightleftharpoons CH_3CHO + H_2$	8	$CH_4 \rightleftharpoons C + 2H_2$
Methane formation		9	$2CO \rightleftharpoons C + CO_2$
4	$2CO + 2H_2 \rightleftharpoons CH_4 + CO_2$	10	$H_2 + CO \rightleftharpoons C + H_2O$
5	$CO + 3H_2 \rightleftharpoons CH_4 + H_2O$	11	$CO_2 + 2H_2 \rightleftharpoons C + 2H_2O$
6	$CH_3CHO \rightleftharpoons CH_4 + CO$		

composition [26–30]. In reality, reaction (1) proceeds according to a complex mechanism, including the formation and decomposition of surface complexes, radicals, the formation of active atoms and their deactivation, but, as is well known, for the thermodynamic analysis, consideration of the intermediate and elementary stages of the reactions does not make sense. Therefore, for first law energy analysis of the thermochemical heat recuperation system in the temperature range from 500 to 1000 K, it was assumed that reaction (1) is principal one and that side reactions do not have a significant effect on the material and energy balance [31,32].

It is well known that the ethanol steam reforming reaction proceeds at moderately high temperatures [34,35]. Therefore, the residual heat of the exhaust gases after the reformer is used to produce steam and ethanol vapor in the steam generator and evaporator, respectively. The synthetic fuel after the reformer consists of different gases, but the main component is hydrogen. The hydrogen-rich fuel is fed into the burner assembly of the fuel-consuming equipment.

Perspectives on the use of TCR can be shown in the following example. The low heat value of ethanol is 29.56 MJ/kg, therefore if heat is generated from ethanol according to the scheme presented in Fig. 2, 29.56 MJ is produced per 1 kg of ethanol. But if the schematic view in Fig. 2b will be considered, there will be another result. The reforming of 1 mol of ethanol (EtOH) by the stoichiometric amount of steam (H_2O) yields 6 mol of hydrogen and 2 mol of carbon dioxide according to reaction (1). At the same time, 374.1 kJ per mole of EtOH (or 8132.9 kJ per kg of EtOH) is absorbed in the reformer. As a

result, the heat of combustion of the reformat products (37688.9 kJ) produced by reforming of 1 kg of EtOH is increased in comparison with the heat of combustion of the initial ethanol (29556.2 kJ). This example shows that the heat of combustion of the initial mixture $EtOH + 3H_2O$ is increased by 8132.9 kJ per 1 kg of EtOH. In addition, the products of ethanol reforming have a moderately high temperature, therefore the amount of heat produced in the fuel-consuming equipment with TCR will be even higher.

The example shown above considers the stoichiometric ethanol-to-steam ratio at the complete ethanol conversion (100%). However, it is known that the enthalpy of reaction (1) and ethanol conversion in an equilibrium condition depends on operational parameters, such as temperature, pressure, and composition of the initial mixture. A change in the reaction enthalpy will lead to a change in the heat balance throughout the TCR system. In addition, the composition of the reforming products also depends on temperature, pressure, and composition, hence the enthalpy of the synthesis gas also varies depending on the operating parameters.

The general goal of this study is the calculation of the heat outputs and inputs of the thermochemical recuperation system that is presented in Fig. 1. Effect of temperature, pressure and ethanol-to-steam ration on the reaction enthalpy and the composition of synthetic fuel in equilibrium can be established by thermodynamic analysis. Therefore, the task of calculating the heat fluxes in the thermochemical heat recuperation system is an adjoint problem with the thermodynamic analysis of the steam ethanol reforming process.

Methodology

The thermodynamic analysis of the ethanol steam reforming process was performed using the software Aspen HYSYS [36]. It should be noted that thermodynamic analysis allows only to establish a functional relationship between the enthalpy of reaction, the composition of the synthesis gas and the control parameters, i.e. parameters that the technologist can within certain limits purposefully change to ensure the optimal mode of the process [13,37]. The control parameters can include the ethanol-to-steam ratio in the feed stream, temperature and pressure. The thermodynamic analysis allows performing very important for designing or optimizing the TCR system calculations, i.e. the determination of the maximum ethanol conversion, reaction enthalpy, and synthesis gas compositions.

The equilibrium composition and reaction enthalpy of the ethanol steam reforming process is calculated by using the Gibbs free energy minimization (GFEM) method, which is based on the fact that the system is thermodynamically favorable when its total Gibbs free energy is at its minimum value and its differential is zero for a given temperature and pressure. The theoretical description of GFEM is given in the previous author's articles [13,15], as well as in the works of other authors [37].

Fig. 2 shows the computational scheme of the thermochemical recuperation process for Aspen HYSYS. The main elements of the scheme completely repeat the elements shown in the schematic diagram (Fig. 1) for the TCR system. Designation of the main elements and flows in Fig. 2: GR-1 – a Gibbs reactor; MIX-1 – a steam-ethanol mixer; HE-1 – boiler (steam generator); HE-2 – an ethanol evaporator; Q1 – the exhaust gas enthalpy (heat) after the fuel-consuming equipment; Q2 – the exhaust gas enthalpy (heat) after the reformer; Q3 – the exhaust gas enthalpy (heat) after the steam generator; Q4 – the exhaust gas enthalpy (heat) after the ethanol evaporator; 1 – liquid water; 2 – liquid ethanol; 3 – steam; 4 – ethanol vapor; 5 – steam-ethanol mixture; 6 – syngas. SET-1,

SET-2 – the user defined functions for the calculation of evaporation heat.

From scheme in Fig. 2, the following heat flows are visible. In the reformer, the heat of exhaust gases is consumed for the ethanol steam reforming reaction and heating of the synthesis gas. Heat flow in the ethanol steam reformer can be expressed as:

$$\Delta Q_{ref} = Q_1 - Q_2 = H_{esr} + \Delta H_{sg}, \quad (2)$$

where H_{esr} – enthalpy of ethanol steam reforming reaction; ΔH_{sg} – enthalpy change of synfuel.

Enthalpy change of synfuel depends on temperature and syngas equilibrium composition, while the equilibrium composition depends on the operating parameters such as temperature, steam-to-ethanol ratio, and pressure. Enthalpy change can be determined as:

$$\Delta H_{sg} = G_5 \left[\sum (\varphi_i \cdot c_{p_i} \cdot T_6) - (h_3 + h_4) \right], \quad (3)$$

where $G_5 = G_3 + G_4 = G_6$ – mass flow through the ethanol steam reformer; φ_i – volume fraction of i th syngas component that defined as the volume of i component divided by the volume of all syngas in pos.6 (in Fig. 2); T_6 – temperature in pos.6; h_3, h_4 – enthalpy of steam (pos.3) and ethanol vapor (pos.4).

Based on thermodynamic analysis, enthalpy of ethanol steam reforming reaction (H_{esr}); volume fraction of each syngas component (φ_i) from expressions (2) and (3) were determined. The source data for thermodynamic calculation was obtained from FactSage database cite[bale2002factsage] [38]. In Appendix A, the $K_{eq,i}$ is the equilibrium constant of i th reaction from Table 1.

The exhaust heat after the fuel-consuming equipment (Q_1) is expressed by:

$$Q_1 = H_{exh} = G_{exh} \cdot c_{p_{exh}} \cdot T_{exh}, \quad (4)$$

where G_{exh} – the mass flow of the exhaust gases, kg/s; T_{exh} – the temperature of the exhaust gases after fuel-consuming

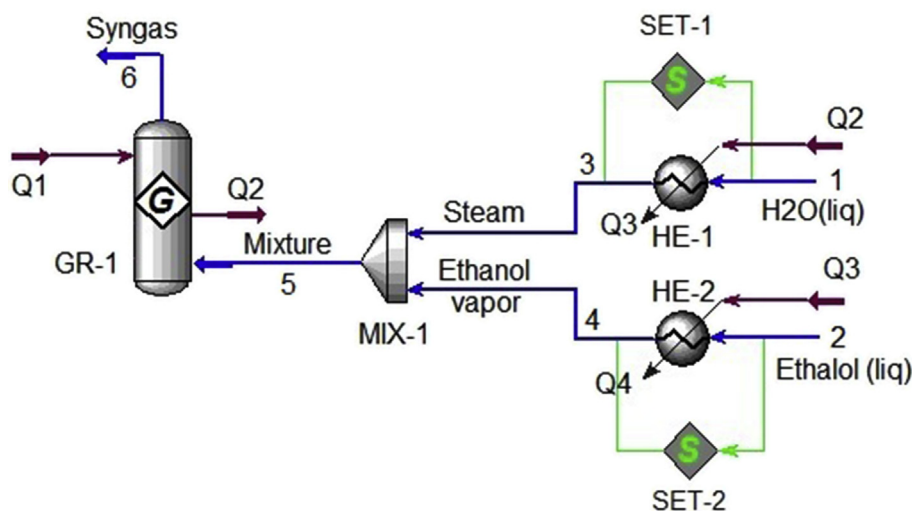


Fig. 2 – The computational scheme of the TCR process for Aspen HYSYS.

equipment, K; $c_{p_{exh}}$ – the heat capacity of the exhaust gas, kJ/(kg·K).

The heat flow in the steam generator (HE-1 in Fig. 2) was determined as follows:

$$\Delta H_3 = Q_2 - Q_3 = G_1 (h'_3 - c_{p(w)} \cdot T_1), \quad (5)$$

where G_1 – the mass flow of liquid water, kg/s; h'_3 – enthalpy of saturated steam at T_3 , J/kg; $c_{p(w)}$ – heat capacity of liquid water, J/(kg·K).

The heat flow in an ethanol evaporator [39] (HE-2 in Fig. 2) was determined similarly to the heat flow in the steam generator.

Schematically, the algorithm of first law energy analysis of thermochemical heat recuperation by ethanol steam reforming is presented in Fig. 3.

Initial conditions and assumptions

The following assumptions have been made in the thermodynamic analysis:

- Steady-state operation.
- Isothermal conditions in the outlet area of reformer are prevailing.
- Pressure drop in the reformer, steam generator, and ethanol evaporator is negligible.
- Plug flow in the Gibbs reactor.
- No interphase and intraparticle mass transfer limitations.
- Exhaust gas composition is fixed.
- No heat loss through the reformer, steam generator, and ethanol evaporator wall.

In addition, in order to compare heat flows at different operating parameters (temperature, pressure, and inlet steam-ethanol composition), the following initial data were taken. The initial data are presented in Table 2. There are in Table 2 T_{exh} – temperature of exhaust gases after fuel-consuming equipment and p_s – saturation pressure.

Table 2 – The initial data for first law energy analysis.

Parameter	Pos. in Fig. 2	Value	Unit
Temperature	1	293	K
	2	293	K
	6	$T_{exh}-150$	K
	GR-1	$T_{exh}-150$	K
	3, 4	$f(p_s)$	K
Mass flow rate	2	1	kg/s
Pressure	GR-1, HE-1, HE-2	equal	Pa

Results and discussion

Results of thermodynamic analysis

As is known, among several possible chemical reactions, the most probable is the reaction with the minimum change of Gibbs free energy. The change of Gibbs free energy that calculated from the data from Table in Appendix A for chemical reactions from Table 1 is shown in Fig. 4. Here in the Fig., the line number corresponds to the reaction number in Table 1.

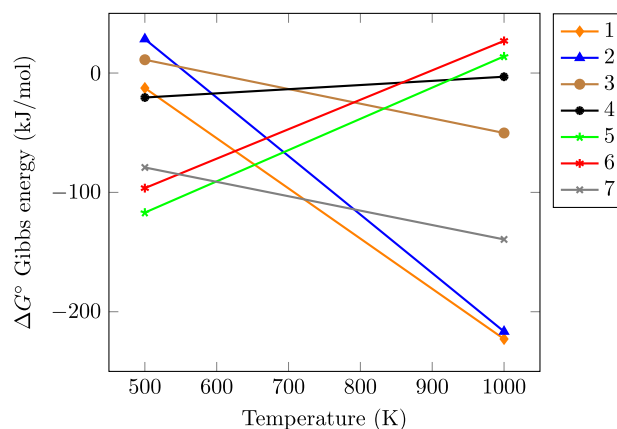


Fig. 4 – The change of Gibbs free energy as a function of temperature for reactions (1–7) from Table 1.

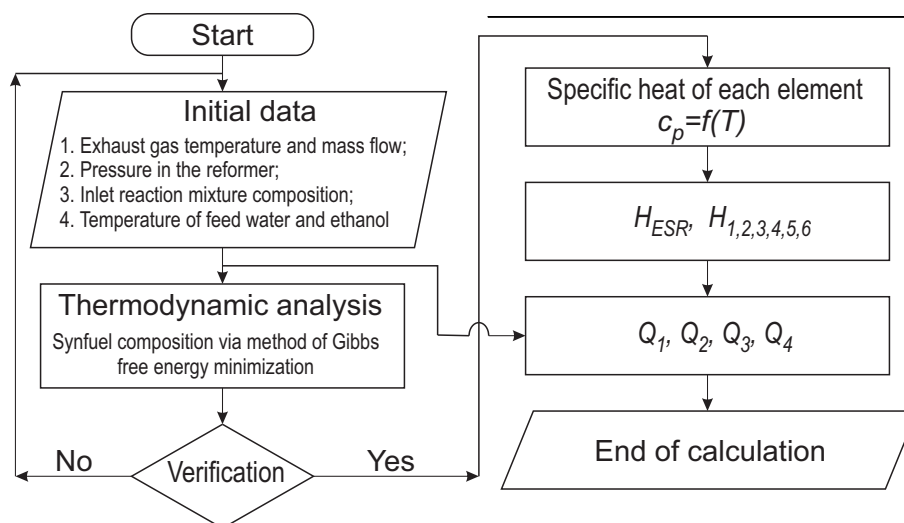


Fig. 3 – The algorithm of first law energy analysis of thermochemical heat recuperation by ethanol steam reforming.

From Fig. 4 that with increasing temperature, the probability of steam ethanol reforming reactions (1) and (2) increases. At temperatures above 700 K, such reactions are predominant, while the change in Gibbs free energy for methanation reactions increases, which reduces the probability of these reactions. However, in the temperature range up to 700 K, the probability of methane formation reactions is high. In particular, the reverse reactions of steam and dry methane reforming [40] in the temperature range up to 600 K are prevalent. In this regard, it is worth expecting that in the temperature range up to 700 K, in addition to hydrogen, a noticeable amount of methane will be contained in the synthesis gas. In this case, in the temperature range above 700 K, the synthesis gas should consist mainly of hydrogen, carbon dioxide, and carbon monoxide. Also, Fig. 4 shows that the ethanol reforming reaction has a direct direction in the temperature range above 500 K. In this regard, it should be expected that the ethanol conversion will tend to 1 at moderately low temperatures.

As mentioned above, the main task of thermodynamic analysis is to determine the enthalpy of ethanol steam reforming process, as well as determining the equilibrium composition of the synthesis gas at the outlet of the reformer. The reaction enthalpy and synthesis gas composition is the basis for first law energy analysis. In this study, the author does not set the main goal to show the results of thermodynamic analysis of the ethanol steam reforming process. Such results are widely represented in the scientific literature [31]. However, it is difficult to generalize and systematize the published results with reference to the task of first law energy analysis of TCR by ethanol steam reforming. The dependencies given below are intended only to show the adequacy of the results obtained in the thermodynamic analysis. For this purpose, the results obtained by the author are compared with the results of the thermodynamic analysis performed by other authors.

In order to verify the results of the thermodynamic analysis, the obtained data were compared with the results of a thermodynamic analysis and experimental investigation of other authors. It should be noted that if in the thermodynamic analysis, the syngas composition does not affect the type and composition of the catalyst, in an experimental study, the composition of the reforming products is affected not only by temperature, pressure, and composition of a steam-ethanol mixture but also by the type of catalyst. This is due to the fact that according to Fig. 4, the probability of many reactions from Table 1 is high in a wide temperature range. The catalyst activates one of the reactions from Table 1, while not affecting the remaining reactions.

Fig. 5 shows a comparison of the results of an experimental study of ethanol steam reforming over a Ni-based catalyst [41] with results of the thermodynamic analysis. Biswas and Kunzru showed that the composition of the catalyst has a significant effect on the synthesis gas composition. The experiments were performed at high residence time (contact time or space velocity) about 4.5g·cath/mol. Such a high time allows us to expect a close to the equilibrium composition of reformat products.

From Fig. 5 that the behaviour of the changing in moles number of the main reformat products in the

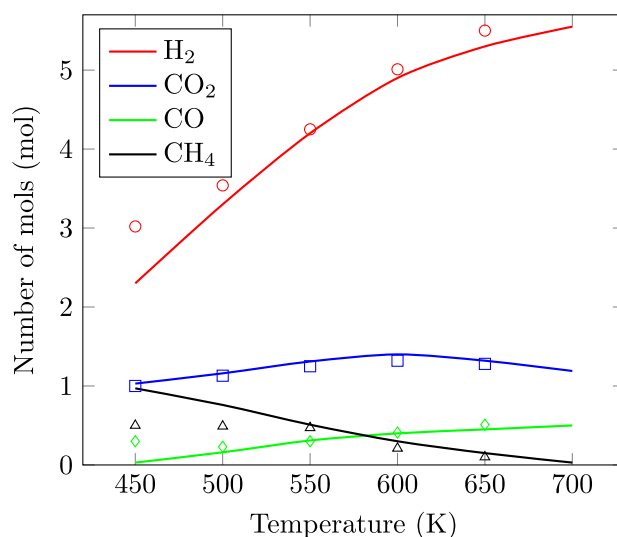


Fig. 5 – A comparison of the results of an experimental study of ethanol steam reforming over a Ni-based catalyst [41] with results of the thermodynamic analysis.

thermodynamic analysis and the experimental study is the same. With increasing temperature, the formation of methane decreases, while the formation of hydrogen increases. This fact agrees well with Fig. 4. Moreover, Fig. 5 presents that methane formation decreasing hydrogen formation when a temperature is lower than 700 K. Conversely, the temperature increase leads to decreasing of methane formation, while hydrogen production is increasing.

In addition, the results of the thermodynamic analysis obtained in this study were compared with the results of the thermodynamic analysis performed by other authors. Fig. 6 shows a comparison of the results obtained by the author and Rossi et al. [42]. The results obtained in this study are in good agreement with the thermodynamic data presented by Rossi et al. [42]. The discrepancy between the results is less than 2–3%. This confirms the obtained results can be used for further first law energy analysis of thermochemical recuperation system by steam ethanol reforming.

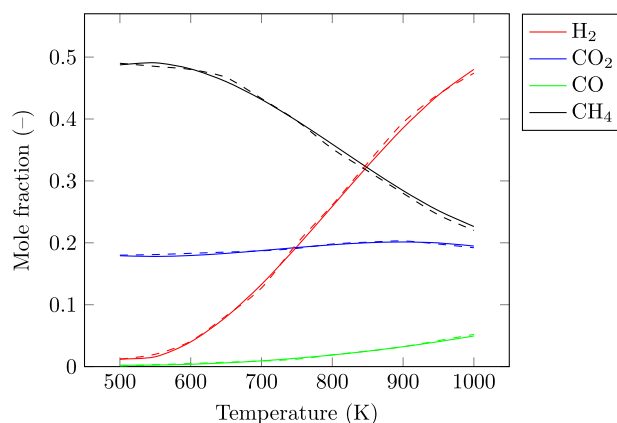


Fig. 6 – Dry syngas composition as a function of temperature at 1 atm for an equimolar steam-to-ethanol ratio: solid lines – previous study; dashed lines – thermodynamic results of Rossi et al. [42].

In addition, Fig. 7 shows the composition of the synthesis gas as a function of temperature for the steam-to-ethanol ratio of 2 and 3. As can be seen from Fig. 7, with increasing temperature the mole fraction of steam in the reaction products decreases when mole fraction of hydrogen increases. This is due to the fact that the change in Gibbs energy for reactions in which steam is consumed and hydrogen is generated decreases. At the same time, Fig. 7 shows that even at high temperatures, residual steam is in the reforming products. For the thermochemical heat recuperation system is a negative fact, since this leads to waste of heat on the steam production in the steam generator (Fig. 2). It should be noted that dependences on Fig. 7b almost completely coincide with the results obtained by Tartakovsky and co-authors [2].

Carbon formation in the ethanol steam reformers is another factor that needs to be considered. However, there is sufficient information in literature that carbon formation can be controlled by improving the coking resistance of the catalysts as well as by regulating the operating conditions such as steam-to-carbon and fuel-to-oxidant ratio [41]. In this study, oxygen-to-carbon ratio in all different feed gas compositions is either equal to or greater than unity, which indicates that there is enough oxygen to oxidize and prevent formation of solid carbon. Hence, carbon formation was omitted from equilibrium expressions.

Heat balance

Based on the results of the thermodynamic analysis, the heat balance of the thermochemical heat recuperation system is compiled. In addition to the source data in Table 2, the following assumptions are made: ethanol vapor and steam after the ethanol evaporator and the steam generator are in a saturation state.

Fig. 8 shows the heat flows for the calculation scheme in Fig. 2. The left column shows the enthalpy of exhaust gases, the right column shows the heat outputs of heat balance: the production of steam and ethanol vapor, synthesis gas heating, the enthalpy of the reforming reaction. Fig. 8 shows that with increasing exhaust gas temperature, the reaction enthalpy increases and, consequently, the amount of heat transferred to the reformer increases. Increasing the steam-to-ethanol ratio increases the heat required for steam production. As the ratio increases from 1 to 3, the heat for steam production

increases threefold. This leads to the fact that for the steam-to-ethanol ratio of 3 there is a heat deficit. In other words, the enthalpy of the exhaust gases is not enough to completely accomplish the steam reforming reaction with the given parameters. Therefore, the system of thermochemical heat recuperation requires either the supply of heat from an external source or a reduction in ethanol consumption.

Fig. 9 shows the heat flow Q_4 in the calculation scheme in Fig. 2. If the heat flux is positive, then the enthalpy of exhaust gases (Q_1) is enough to perform thermochemical recuperation at the given parameters. If the heat flow Q_1 is negative, then either the supply of heat from an external source or a reduction in ethanol consumption is necessary. Fig. 9a shows that thermochemical recuperation by ethanol steam reforming at a steam-to-ethanol ratio of 3 is energetically impractical in the temperature range from 500 to 1000 K since requires the supply of heat from an external source.

Figs. 8 and 9a shows the heat flows for pressure 1 bar in the reformer, the steam generator, and the ethanol evaporator. However, the change in pressure affects heat flow. The increase in pressure leads, firstly, to a change in the composition of the synthesis gas; secondly, to a change in the reaction enthalpy; thirdly, to an increase in the heat required for the production of steam and ethanol vapor. Fig. 9b shows heat flow Q_1 for different pressure at the steam-to-ethanol ratio of 3.

Fig. 9b shows that an increase in pressure increases the heat deficit in the thermochemical heat recuperation system. However, the effect of pressure in the range from 800 to 900 K is negligible. This is due to the fact that with increasing pressure, although the heat consumption for the production of steam and ethanol vapor increases, the reaction enthalpy decreases and, consequently, the heat consumption in the reformer decreases. In the temperature range from 500 to 700 K and above 900 K, the reaction enthalpy slightly depends on the pressure and the heat flow Q_1 is mainly influenced by the heat consumption on the production of steam and ethanol vapor.

Recuperation rate

The efficiency of application the TCR systems by ethanol steam reforming is depending on the heat transferred in the reformer for the ethanol reforming reaction and the syngas

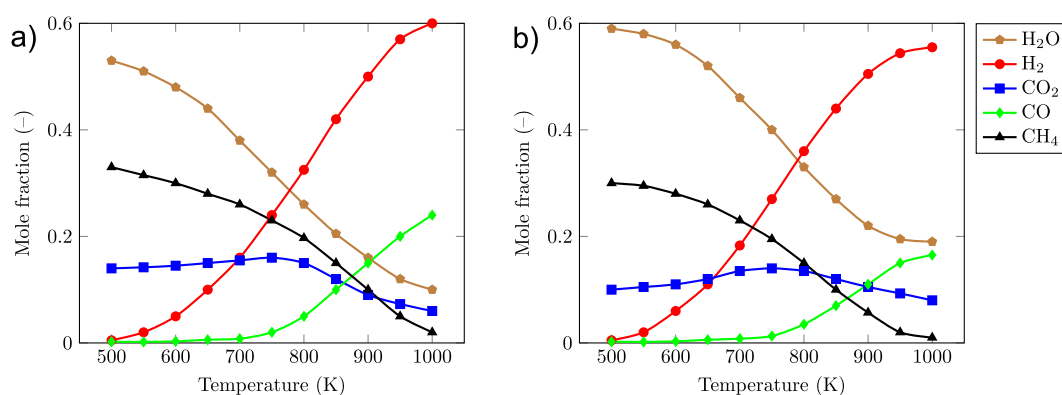


Fig. 7 – Syngas composition as a function of temperature for steam-to-ethanol ratio of 2 (a) and 3 (b).

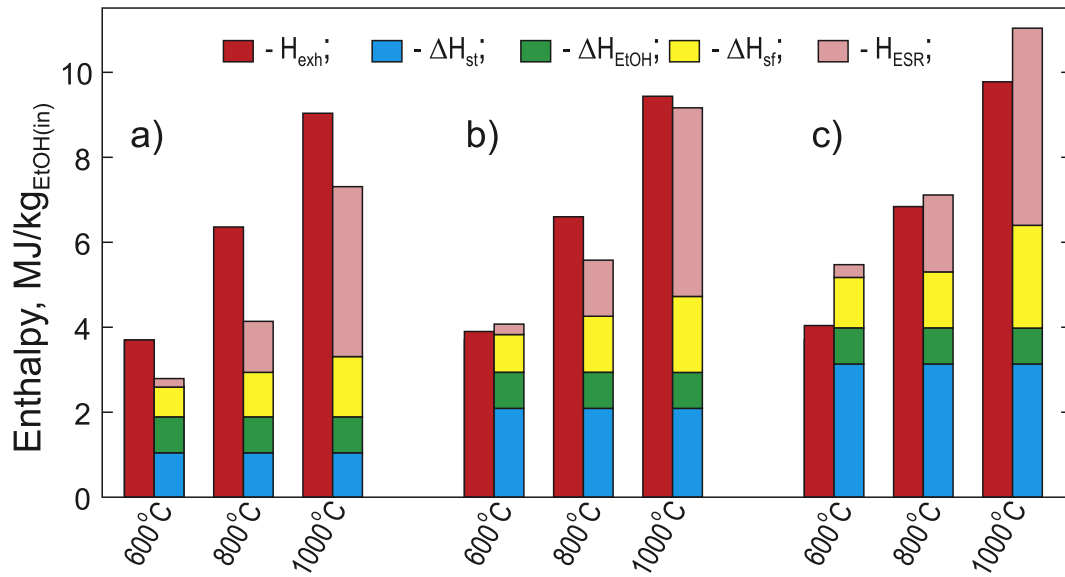


Fig. 8 – Heat balance for TCR system for various steam-to-ethanol ratio: a) 1; b) 2; c) 3.

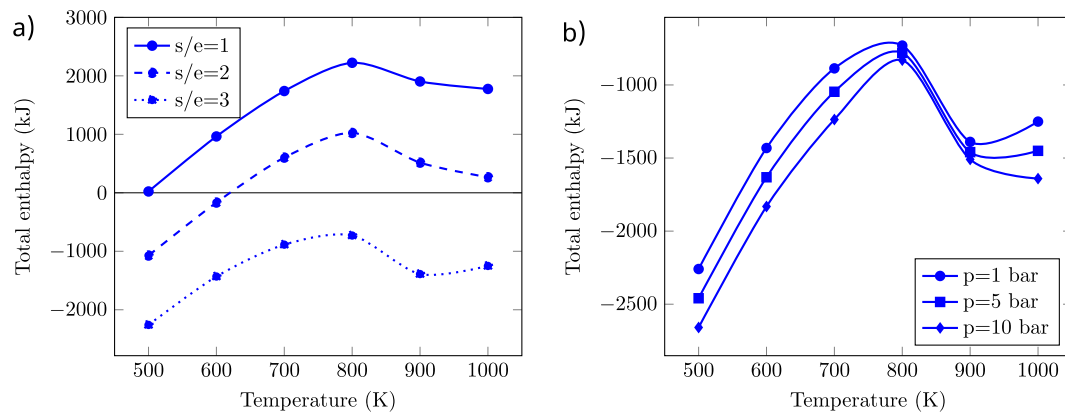


Fig. 9 – Heat flow Q4 for scheme in Fig. 2 as a function of temperature: a) for various steam-to-ethanol ratio at $p = 1$ bar; b) for various pressure at $s/e = 3$.

heating. If ethanol steam reforming doesn't occur in the reformer, then the efficiency of using of TCR system equals to zero or even has a negative effect, due to the fact that exhaust heat is used for generation of steam and ethanol vapor. In addition, an increasing steam-to-ethanol ratio in feed stream leads to an increase of mole fraction of non-combustible species in the syngas, in particular steam. This fact can be proved from Fig. 7. Residual steam is a component of syngas but from the energy point of view, this residual steam doesn't play a positive role in the combustion process. Thus, residual steam is non-combustible inert material of the synfuel. Moreover, a large amount of heat is required for steam generation and from an energy point of view, a large molar fraction of steam in the synfuel is negative. To determine the optimal steam-to-ethanol ratio for the TCR system the recovery rate in the reformer is calculated and analyzed.

According to definition, the recovery rate can be expressed as follows:

$$R = \frac{\Delta H_{rec}}{H_{exh}}, \quad (6)$$

where ΔH_{rec} – recuperated heat in the TCR system, kJ; H_{exh} – exhaust heat, kJ.

According to Fig. 2, the exhaust gases are used for the ethanol steam reforming reaction and for heating of the syngas. However, the syngas consists of combustible elements (H_2 , CO , CH_4) and non-combustible elements (H_2O , CO_2). Therefore, the enthalpy of noncombustible elements in the syngas can't be taken into account as a recovered heat because in this case, these elements are the inert materials. Therefore, the recovered heat in the thermochemical recuperation system can be calculated as follows:

$$\Delta H_{rec} = H_{SER} + H_{SF} - H_{H_2O} - H_{CO_2} - H_{steam} - H_{EtOH}, \quad (7)$$

where H_{SER} – enthalpy of steam ethanol reforming reaction, kJ; H_{SF} – the synfuel enthalpy, kJ; H_{H_2O} , H_{CO_2} – enthalpy of steam and carbon dioxide that are contained in the syngas, kJ; H_{steam} – enthalpy of steam before the reformer, kJ; H_{EtOH} – enthalpy of ethanol vapor before the reformer, kJ.

Fig. 10 shows the dependence of a recovery rate as a function of temperature for different steam-to-ethanol ratio

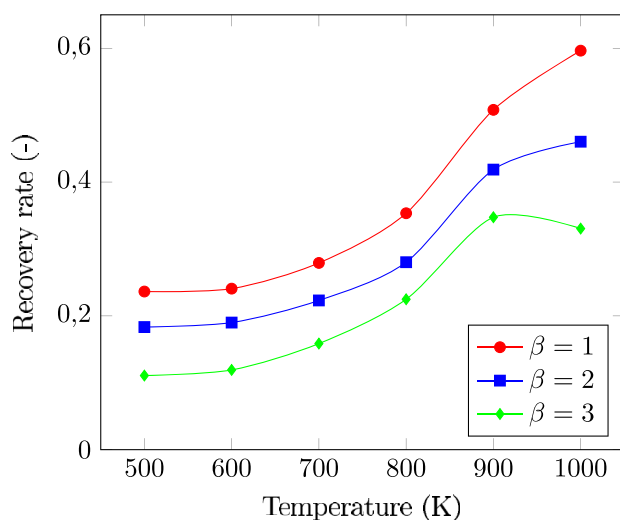


Fig. 10 – Recovery rate as a function of temperature for different steam-to-ethanol ratio at $p = 5$ bar.

at $p = 5$ bar. From Fig. 10 that the maximum recovery rate in the reformer takes place at a steam-to-ethanol ratio equal to 1. This is due to the fact that at this ratio the main amount of heat is transferred to the combustible components, while the volume of non-combustible components in the synthesis gas is minimal. For a ratio of 2 and 3 at temperature of 900 K, a gradual increase in recovery rate is observed. However, in the temperature range above 900 K, an increase in the recovery rate decreases, and for ratio 3 it even decreases. This is due to the fact that the reaction enthalpy at this temperature reaches close to the maximum value and the heat flow in the reformer is spent mainly on heating the synthesis gas. At the same time in the synthesis gas contains a large amount of residual steam. This leads to a decrease in the recovery rate.

Based on the dependencies in Fig. 10 it can be concluded that from the energetic point of view, the optimal ratio is 1, since in this case, the maximum recovery rate is achieved. However, when developing the real TCR systems by steam ethanol reforming, operational efficiency must be taken into account. For example, it is known that carbon deposition may occur at low steam-to-ethanol ratios in the reformer, which decreases the efficiency of the reforming process. Therefore, the determination of the optimal parameters of the TCR system from the point of view of operational characteristics is an

important engineering task and it will be considered in future publications.

Conclusion

The thermochemical heat recuperation system by steam ethanol reforming is considered. To determine the effect of operating parameters such as temperature, pressure and steam-to-ethanol ratio on the heat inputs and outputs, first law energy analysis was performed. First law energy analysis is based on thermodynamic analysis of the TCR system by Gibbs free energy minimization method. Based on the results of the thermodynamic analysis, the heat balance of the thermochemical heat recuperation system is compiled. It was found that for a high steam-to-ethanol ratio (above 3), to perform thermochemical recuperation an external heat must be supplied to the TCR system. The heat deficit for steam-to-ethanol ratio 3 is from 1 to 2 MJ/kg_{EtOH} in the temperature range from 500 to 1000 K. A pressure in the range from 1 to 10 bar does not have a noticeable effect on the heat balance of the TCR system. This is due to the fact that with increasing pressure, although the heat consumption for the production of steam and ethanol vapor increases, the reaction enthalpy decreases and, consequently, the heat consumption in the reformer decreases.

An expression for determining the recovery rate for the thermochemical recuperation system is obtained and justified in this article. The maximum recovery rate in the reformer takes place at a steam-to-ethanol ratio equal to 1. For a steam-to-ethanol ratio of 2 and 3 at temperature of 900 K, a gradual increase in the recovery rate is observed. However, in the temperature range above 900 K, an increase in the recovery rate decreases, and for ratio 3 it even decreases.

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Appendix A

The equilibrium constants for the thermodynamic analysis from FactSage Database [38].

T(K)	$K_{eq,1}$	$K_{eq,2}$	$K_{eq,3}$	$K_{eq,4}$	$K_{eq,5}$	$K_{eq,6}$	$K_{eq,7}$
500.00	2.06e+01	1.06e-03	6.84e-02	1.39e+02	1.67e+12	1.20e+10	1.86e+08
550.00	1.32e+03	3.85e-01	3.31e-01	5.85e+01	6.33e+09	1.08e+08	1.25e+08
600.00	4.45e+04	5.40e+01	1.23e+00	2.86e+01	5.93e+07	2.06e+06	9.02e+07
650.00	9.06e+05	3.62e+03	3.784e+00	1.58e+01	1.12e+06	7.10e+04	6.80e+07
700.00	1.23e+07	1.35e+05	9.90e+00	9.55e+00	3.70e+04	3.88e+03	5.31e+07
750.00	1.22e+08	3.16e+06	2.27e+01	6.20e+00	1.91e+03	3.08e+02	4.28e+07
800.00	9.22e+08	5.03e+07	4.72e+01	4.28e+00	1.42e+02	3.32e+01	3.53e+07
850.00	5.57e+09	5.80e+08	9.00e+01	3.09e+00	1.42e+01	4.61e+00	2.97e+07
900.00	2.79e+10	5.12e+09	1.59e+02	2.33e+00	1.84e+00	7.92e-01	2.54e+07
950.00	1.19e+11	3.60e+10	2.66e+02	1.81e+00	2.96e-01	1.63e-01	2.20e+07
1000.00	4.43e+11	2.09e+11	4.22e+02	1.45e+00	1.45e+00	3.91e-02	1.93e+07

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijhydene.2019.10.009>.

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